

```
In [29]: import pandas as pd
```

```
In [30]: pd.__version__
```

```
Out[30]: '2.3.3'
```

```
In [31]: df = pd.read_csv(r'C:\Users\pandi\Desktop\Abhiraj\Data Science\Notes\Learning\10
```

```
In [32]: df
```

```
Out[32]:
```

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [33]: type(df)
```

```
Out[33]: pandas.core.frame.DataFrame
```

```
In [34]: len(df)
```

```
Out[34]: 195
```

```
In [35]: df.columns
```

```
Out[35]: Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
               'IncomeGroup'],
              dtype='object')
```

```
In [36]: len(df.columns)
```

Out[36]: 5

```
In [37]: df.shape
```

Out[37]: (195, 5)

```
In [38]: df.isnull()
```

Out[38]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

```
In [39]: df.isnull().sum()
```

Out[39]: CountryName 0
CountryCode 0
BirthRate 0
InternetUsers 0
IncomeGroup 0
dtype: int64

```
In [40]: df.head() # top 5 rows we will get by default
```

Out[40]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

```
In [41]: df.head(2) #top 2 rows
```

Out[41]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

In [42]: `df.tail()` *# by default we will get last 5 rows*

Out[42]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

In [43]: `df.dtypes` *#it will give the data types of the dataframe*

Out[43]:

```
CountryName      object
CountryCode      object
BirthRate        float64
InternetUsers     float64
IncomeGroup      object
dtype: object
```

In [44]: `df.info()` *#by applying info all text data marked as categorie and number wil*

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CountryName     195 non-null   object
1   CountryCode     195 non-null   object
2   BirthRate       195 non-null   float64
3   InternetUsers   195 non-null   float64
4   IncomeGroup     195 non-null   object
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
```

In [45]: `df`

Out[45]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

Slicing In Pandas

In [46]: df[:]

Out[46]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [47]: `df[:6]`

Out[47]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income

In [49]: `df[3]` *# it will not work*

```

-----
KeyError                                Traceback (most recent call last)
File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3812, in Index.get_loc(self, key)
    3811 try:
-> 3812     return self._engine.get_loc(casted_key)
    3813 except KeyError as err:

File pandas\_libs\index.pyx:167, in pandas._libs.index.IndexEngine.get_loc()

File pandas\_libs\index.pyx:196, in pandas._libs.index.IndexEngine.get_loc()

File pandas\_libs\hashtable_class_helper.pxi:7088, in pandas._libs.hashtable.PyObjectHashTable.get_item()

File pandas\_libs\hashtable_class_helper.pxi:7096, in pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: 3

```

The above exception was the direct cause of the following exception:

```

KeyError                                Traceback (most recent call last)
Cell In[49], line 1
----> 1 df[3] # it will not work

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:4113, in DataFrame.__getitem__(self, key)
    4111 if self.columns.nlevels > 1:
    4112     return self._getitem_multilevel(key)
-> 4113 indexer = self.columns.get_loc(key)
    4114 if is_integer(indexer):
    4115     indexer = [indexer]

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3819, in Index.get_loc(self, key)
    3814 if isinstance(casted_key, slice) or (
    3815     isinstance(casted_key, abc.Iterable)
    3816     and any(isinstance(x, slice) for x in casted_key)
    3817 ):
    3818     raise InvalidIndexError(key)
-> 3819 raise KeyError(key) from err
    3820 except TypeError:
    3821     # If we have a listlike key, _check_indexing_error will raise
    3822     # InvalidIndexError. Otherwise we fall through and re-raise
    3823     # the TypeError.
    3824     self._check_indexing_error(key)

KeyError: 3

```

In [50]: df[3:]

Out[50]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income
6	Armenia	ARM	13.308	41.9	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

192 rows × 5 columns

In []: df[0:50:5]

In [51]: df[:,10]

Out[51]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.900000	High income
10	Azerbaijan	AZE	18.300	58.700000	Upper middle income
20	Belarus	BLR	12.500	54.170000	Upper middle income
30	Canada	CAN	10.900	85.800000	High income
40	Costa Rica	CRI	15.022	45.960000	Upper middle income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
60	Gabon	GAB	30.555	9.200000	Upper middle income
70	Greenland	GRL	14.500	65.800000	High income
80	India	IND	20.291	15.100000	Lower middle income
90	Kazakhstan	KAZ	22.730	54.000000	Upper middle income
100	Libya	LBY	21.425	16.500000	Upper middle income
110	Moldova	MDA	12.141	45.000000	Lower middle income
120	Mozambique	MOZ	39.705	5.400000	Low income
130	Netherlands	NLD	10.200	93.956400	High income
140	Poland	POL	9.600	62.849200	High income
150	Sudan	SDN	33.477	22.700000	Lower middle income
160	Suriname	SUR	18.455	37.400000	Upper middle income
170	Tajikistan	TJK	30.792	16.000000	Lower middle income
180	Uruguay	URY	14.374	57.690000	High income
190	Yemen, Rep.	YEM	32.947	20.000000	Lower middle income

In [52]: df

Out[52]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [53]: `df[::10]`

Out[53]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5000	Low income
184	Venezuela, RB	VEN	19.842	54.9000	High income
174	Trinidad and Tobago	TTO	14.590	63.8000	High income
164	Swaziland	SWZ	30.093	24.7000	Lower middle income
154	Sierra Leone	SLE	36.729	1.7000	Low income
144	French Polynesia	PYF	16.393	56.8000	High income
134	Oman	OMN	20.419	66.4500	High income
124	Malaysia	MYS	16.805	66.9700	Upper middle income
114	Macedonia, FYR	MKD	11.222	65.2400	Upper middle income
104	Lesotho	LSO	28.738	5.0000	Lower middle income
94	Kiribati	KIR	29.044	11.5000	Lower middle income
84	Iceland	ISL	13.400	96.5468	High income
74	Hong Kong SAR, China	HKG	7.900	74.2000	High income
64	Guinea	GIN	37.337	1.6000	Low income
54	Estonia	EST	10.300	79.4000	High income
44	Czech Republic	CZE	10.200	74.1104	High income
34	Cote d'Ivoire	CIV	37.320	8.4000	Lower middle income
24	Brazil	BRA	14.931	51.0400	Upper middle income
14	Burkina Faso	BFA	40.551	9.1000	Low income
4	United Arab Emirates	ARE	11.044	88.0000	High income

In [54]: `df[:: -1]`

Out[54]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
191	South Africa	ZAF	20.850	46.5	Upper middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
...	...	...	...	...	...
4	United Arab Emirates	ARE	11.044	88.0	High income
3	Albania	ALB	12.877	57.2	Upper middle income
2	Angola	AGO	45.985	19.1	Upper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

195 rows × 5 columns

In [56]: `df.describe()` *# it is referring for descriptive stats /give information about o*

Out[56]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [55]: `# splitting the data`
`df.head()`

Out[55]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

```
In [57]: df_cat = df[['CountryName', 'CountryCode', 'IncomeGroup']]
df_cat
```

Out[57]:

	CountryName	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income
...	...	...	...
190	Yemen, Rep.	YEM	Lower middle income
191	South Africa	ZAF	Upper middle income
192	Congo, Dem. Rep.	COD	Low income
193	Zambia	ZMB	Lower middle income
194	Zimbabwe	ZWE	Low income

195 rows × 3 columns

```
In [58]: df_num = df[['BirthRate', 'InternetUsers']]
df_num
```

Out[58]:

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985	19.1
3	12.877	57.2
4	11.044	88.0
...	...	...
190	32.947	20.0
191	20.850	46.5
192	42.394	2.2
193	40.471	15.4
194	35.715	18.5

195 rows × 2 columns

```
In [59]: print(df.shape)
print(df_cat.shape)
print(df_num.shape)
```

(195, 5)

(195, 3)

(195, 2)

```
In [60]: print(df.columns)
print(df_cat.columns)
print(df_num.columns)
```

Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
'IncomeGroup'],
dtype='object')

Index(['CountryName', 'CountryCode', 'IncomeGroup'], dtype='object')

Index(['BirthRate', 'InternetUsers'], dtype='object')

```
In [63]: df_cat.describe() # min and max value is missing
```

Out[63]:

	CountryName	CountryCode	IncomeGroup
count	195	195	195
unique	195	195	4
top	Aruba	ABW	High income
freq	1	1	67

```
In [62]: df_num.describe()
```

Out[62]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [64]: `df_cat.describe().T` *# transpose of the output*

Out[64]:

	count	unique	top	freq
CountryName	195	195	Aruba	1
CountryCode	195	195	ABW	1
IncomeGroup	195	4	High income	67

In [65]: `df_num.describe().T`

Out[65]:

	count	mean	std	min	25%	50%	75%	max
BirthRate	195.0	21.469928	10.605467	7.9	12.1205	19.68	29.7595	49.6610
InternetUsers	195.0	42.076471	29.030788	0.9	14.5200	41.00	66.2250	96.5468

In [66]: `df_cat`
`df_cat.columns = ['A', 'B', 'C']` *#you can change the attribute name but more tha*
`df_cat`

Out[66]:

	A	B	C
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income
...	...	...	...
190	Yemen, Rep.	YEM	Lower middle income
191	South Africa	ZAF	Upper middle income
192	Congo, Dem. Rep.	COD	Low income
193	Zambia	ZMB	Lower middle income
194	Zimbabwe	ZWE	Low income

195 rows × 3 columns

```
In [ ]: df_cat.columns = ['CountryName', 'CountryCode', 'IncomeGroup']
df_cat
```

Condition in Pandas

In [67]:

df

Out[67]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [68]: `df['InternetUsers'] < 2`

Out[68]:

```

0      False
1      False
2      False
3      False
4      False
...
190    False
191    False
192    False
193    False
194    False
Name: InternetUsers, Length: 195, dtype: bool

```

In [69]: `df[df['InternetUsers'] < 2]`

Out[69]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
52	Eritrea	ERI	34.800	0.9	Low income
55	Ethiopia	ETH	32.925	1.9	Low income
64	Guinea	GIN	37.337	1.6	Low income
117	Myanmar	MMR	18.119	1.6	Lower middle income
127	Niger	NER	49.661	1.7	Low income
154	Sierra Leone	SLE	36.729	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income
172	Timor-Leste	TLS	35.755	1.1	Lower middle income

In [70]: `len(df[df['InternetUsers'] < 2])`

Out[70]: 9

In [71]: `df[df['BirthRate'] > 40]`

Out[71]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.1	Upper middle income
11	Burundi	BDI	44.151	1.3	Low income
14	Burkina Faso	BFA	40.551	9.1	Low income
65	Gambia, The	GMB	42.525	14.0	Low income
115	Mali	MLI	44.138	3.5	Low income
127	Niger	NER	49.661	1.7	Low income
128	Nigeria	NGA	40.045	38.0	Lower middle income
156	Somalia	SOM	43.891	1.5	Low income
167	Chad	TCD	45.745	2.3	Low income
178	Uganda	UGA	43.474	16.2	Low income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income

In [72]: `len(df[df['BirthRate'] > 40])`

Out[72]: 12

In [73]: `df[(df.BirthRate > 40) & (df.InternetUsers < 2)] # applying more than one filter`

Out[73]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
127	Niger	NER	49.661	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income

In [88]: df

Out[88]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [74]: `df['IncomeGroup'].unique()` *# with the help of this function we can get the dis*

Out[74]: `array(['High income', 'Low income', 'Upper middle income', 'Lower middle income'], dtype=object)`

In [75]: `df['IncomeGroup'].nunique()` *# we can get distinct count of the attribute*

Out[75]: 4

Vizualization

In [76]: `import matplotlib.pyplot as plt`
`import seaborn as sns`
`%matplotlib inline`
`plt.rcParams['figure.figsize'] = 6,2`

`import warnings`
`warnings.filterwarnings('ignore')`

```
In [77]: df['InternetUsers']
```

```
Out[77]: 0      78.9
         1       5.9
         2      19.1
         3      57.2
         4      88.0
         ...
        190     20.0
        191     46.5
        192       2.2
        193     15.4
        194     18.5
        Name: InternetUsers, Length: 195, dtype: float64
```

```
In [78]: vis1 = sns.distplot(df['InternetUsers']) # Distribution plot
        #vis1 = sns.histplot(df['InternetUsers']) # Distribution plot
```

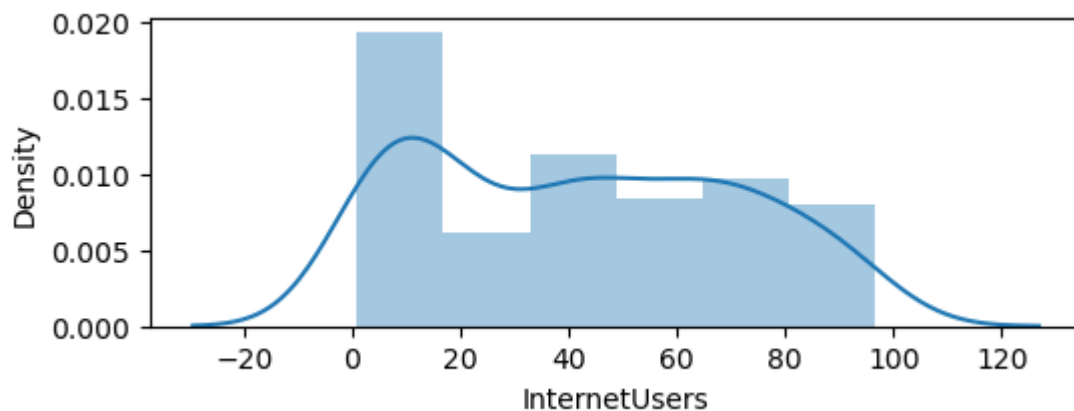
C:\Users\pandi\AppData\Local\Temp\ipykernel_8628\1057093154.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

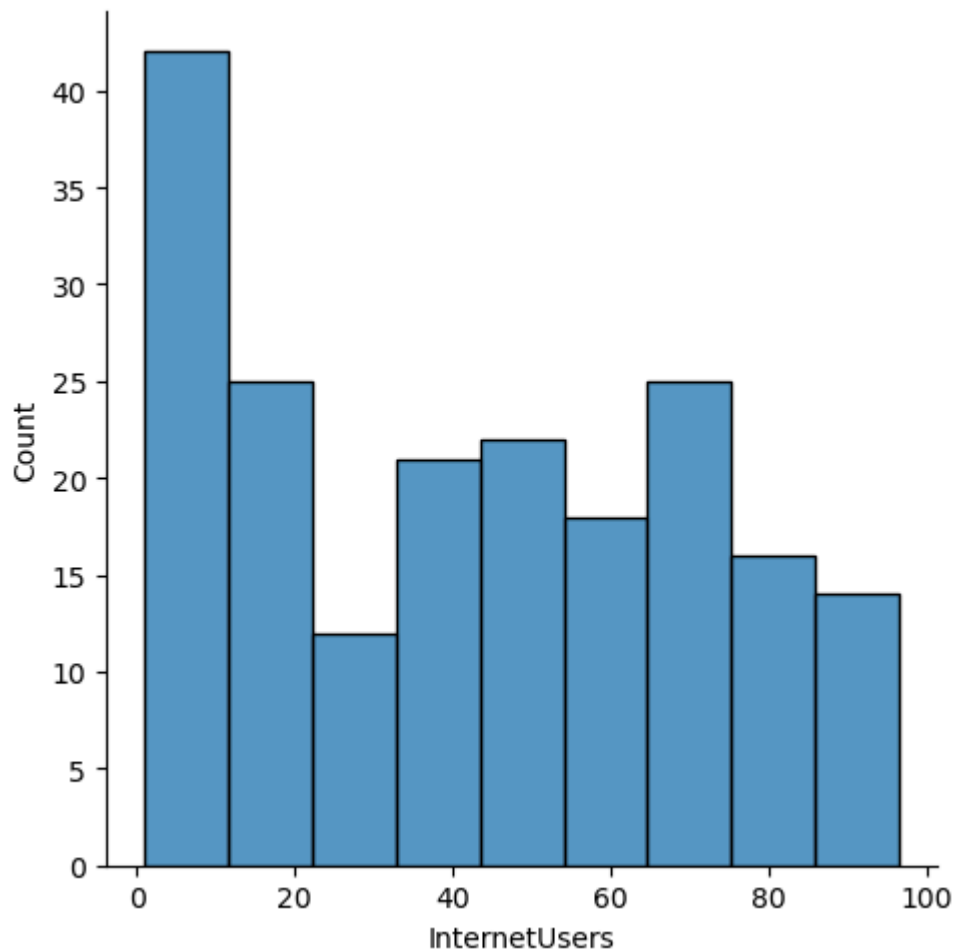
Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
vis1 = sns.distplot(df['InternetUsers']) # Distribution plot
```



```
In [79]: vis1 = sns.displot(df['InternetUsers'])
```



In [86]: `vis1 = sns.distplot(df['InternetUsers'], bins = 10) #bins describe the bar corner`

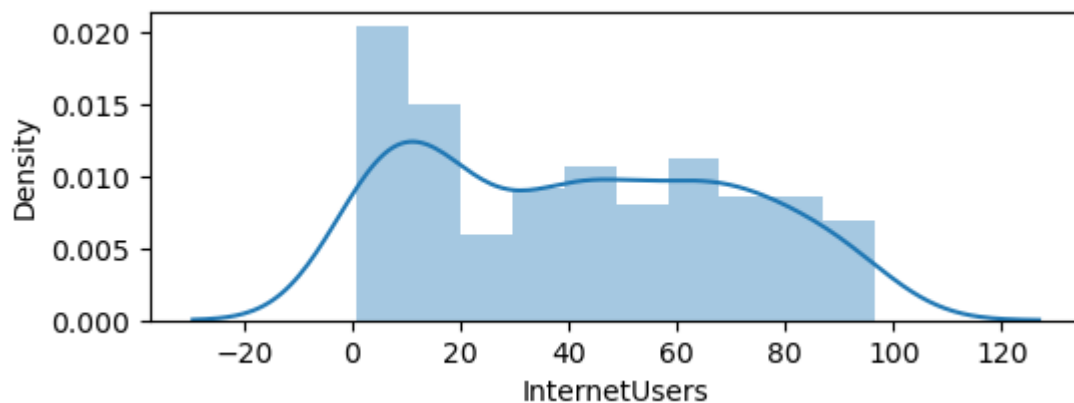
C:\Users\pandi\AppData\Local\Temp\ipykernel_8628\88551027.py:1: UserWarning:

'distplot' is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either 'displot' (a figure-level function with similar flexibility) or 'histplot' (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

`vis1 = sns.distplot(df['InternetUsers'], bins = 10) #bins describe the bar corner`
 touchpoints are chnage



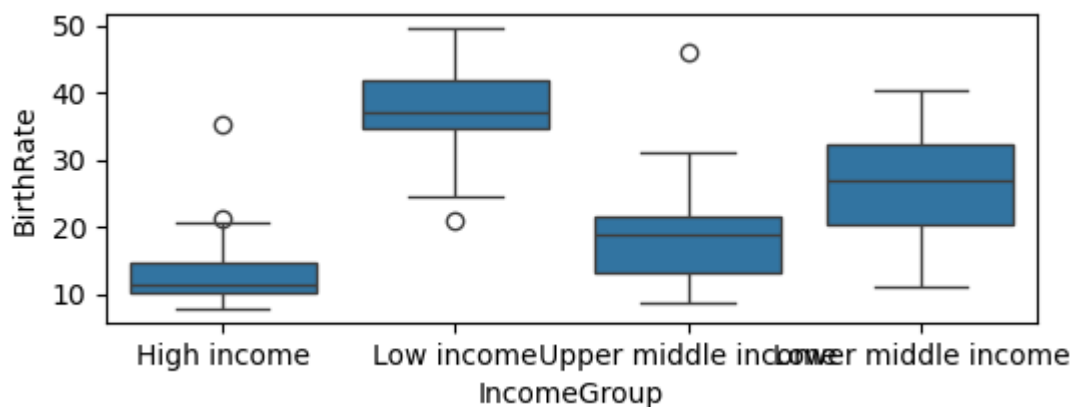
In [90]: `df`

Out[90]:

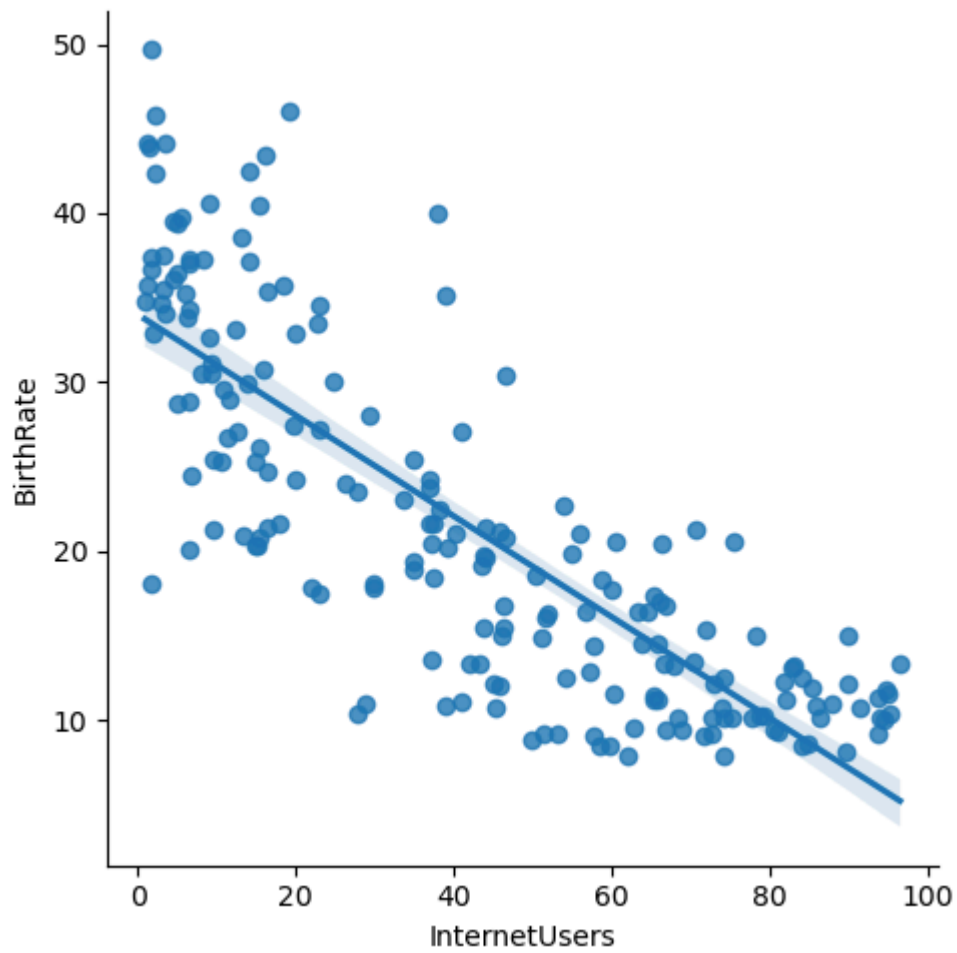
	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

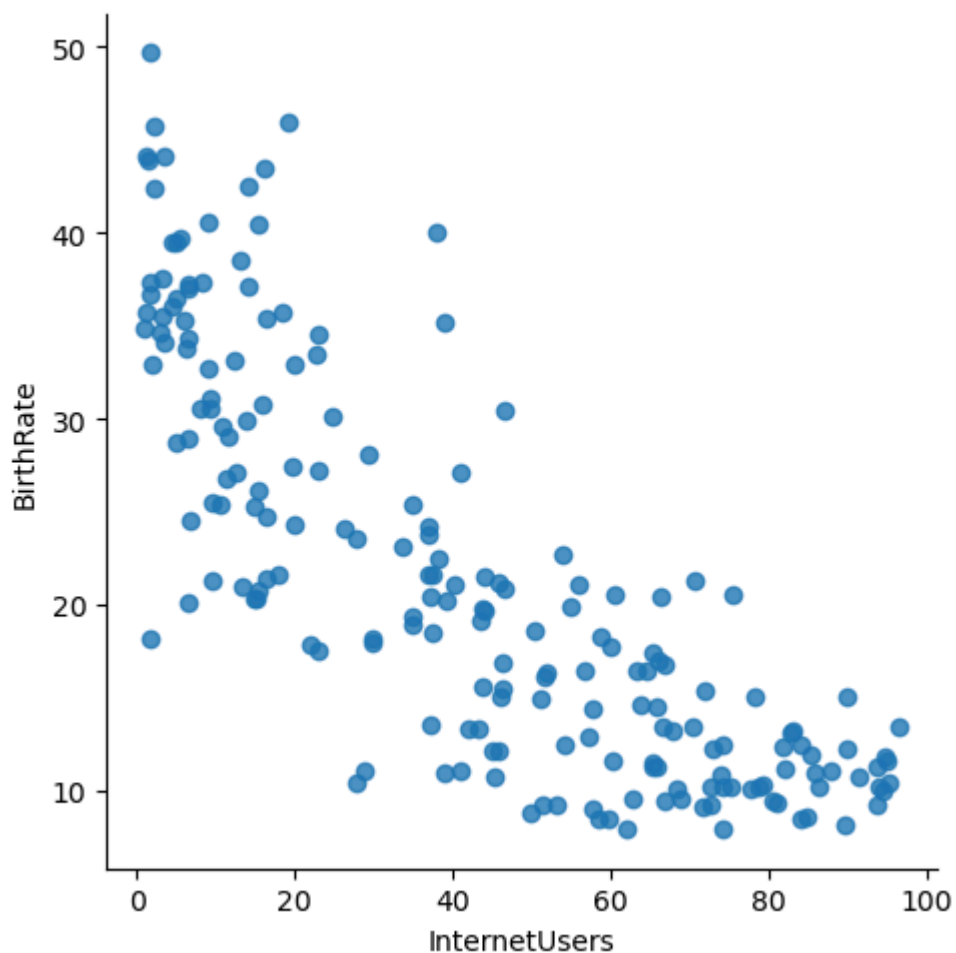
In [94]: `#Box Plot`
`vis4 = sns.boxplot(data=df, x= 'IncomeGroup' , y = 'BirthRate' ) # bi-variate`



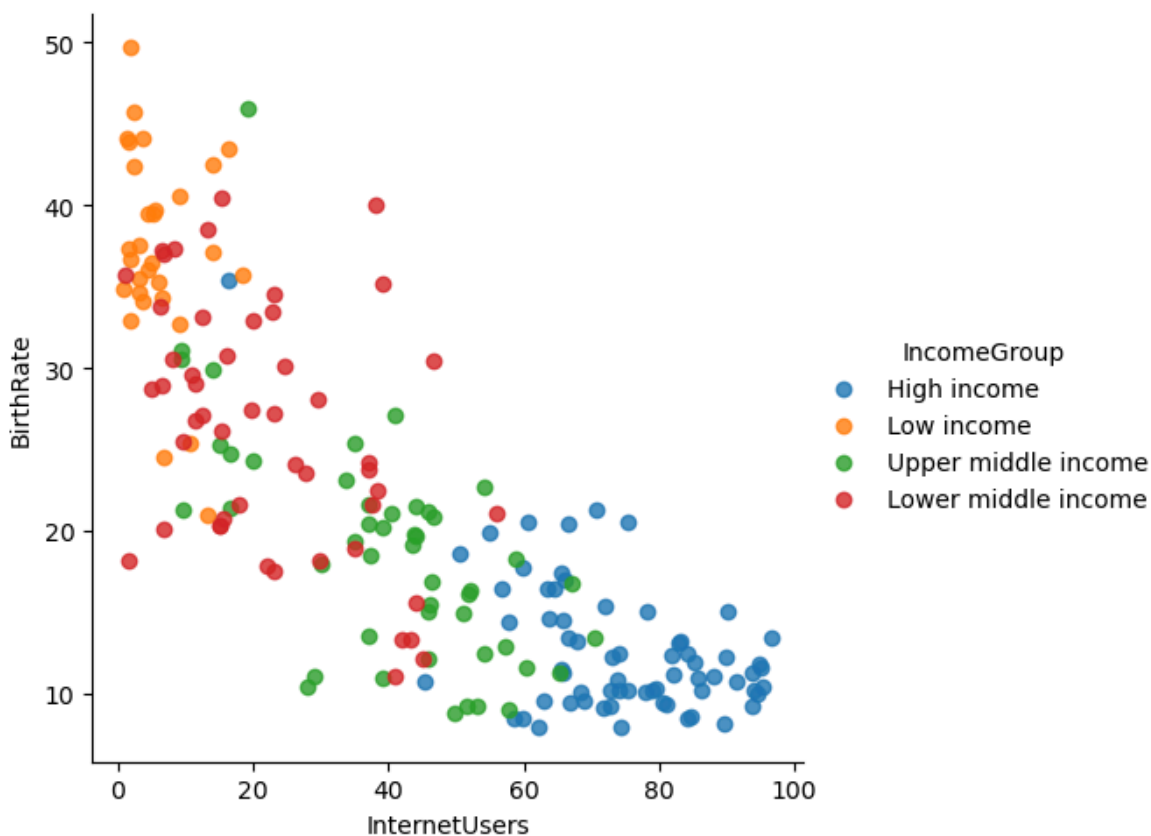
In [95]: `vis5 = sns.lmplot(data= df , x = 'InternetUsers' , y = 'BirthRate')`



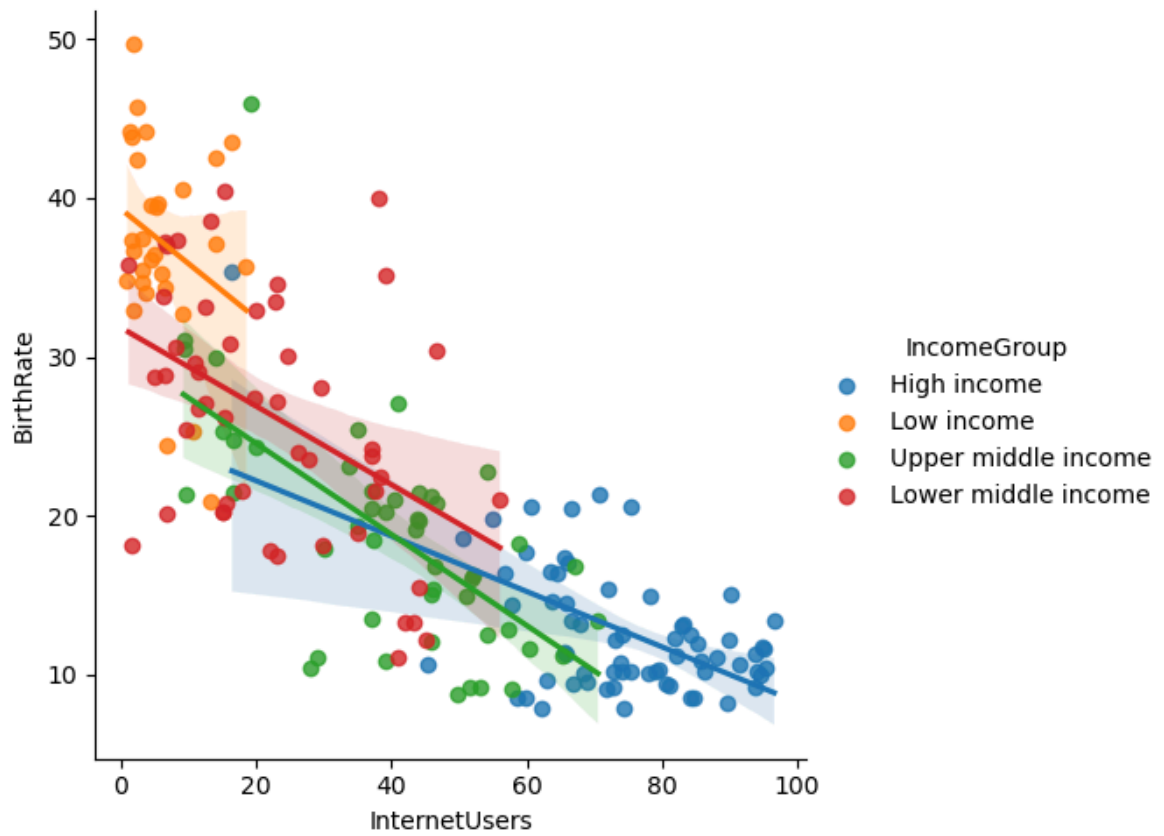
```
In [97]: vis5 = sns.lmplot(data= df , x = 'InternetUsers' , y = 'BirthRate',fit_reg = Fal
```



```
In [98]: vis5 = sns.lmplot(data= df , x = 'InternetUsers' , y = 'BirthRate',fit_reg = Fal
```



```
In [100... vis5 = sns.lmplot(data= df , x = 'InternetUsers' , y = 'BirthRate',fit_reg = Tru
```



In []:

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In []:

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